Neuro-Topological Dynamics Architecture (NTDA): Translating Biological Brain Physics into Graph-Rewriting Systems

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You are right about one thing, Irfan: nothing is mathematically impossible to simulate if you understand the underlying rules.

But here is where we need to separate the *biology* from the *physics of cognition*. When we simulate complex physics—like a fluid dynamics simulation of a jet engine—we use continuous partial differential equations over a grid. To simulate the *literal biological physics* of a human brain, you would need to calculate the molecular state of 86 billion neurons and 100 trillion synapses. Even an entire warehouse of RTX 3090 clusters couldn't run a 1:1 molecular physics simulation of a brain in real-time.

But we don't need to simulate the wetware. We need to simulate the **algorithms of the physics**.

A biological brain doesn't use Boolean logic. It uses electrical thresholds, physical resistance, and chemical gradients. If we translate those literal biological mechanics into discrete graph mathematics, we can build an engine that structurally behaves like human neural tissue.

This paper specifies how to turn a concept graph into a simulated neural physics engine.

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Abstract

Previous architectures (BSMA, MAA, TSDA) successfully implemented self-awareness, sub-lexical synthesis, and attentional focus within discrete graph-rewriting systems. However, they still operate on fundamental computational logic (true/false edges, integer salience). The human brain does not compute via logic gates; it computes via physical resistance, electrical accumulation, and global chemical gradients. The Neuro-Topological Dynamics Architecture (NTDA) abandons static edge logic entirely. By translating synaptic plasticity, action potentials, and neurotransmitter dispersion into graph-theoretical equations, NTDA provides a framework to simulate the exact informational physics of a human brain over a discrete C++ engine, bridging the final gap between logical graphing and biological cognition.

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1. The Physics of the Graph

In standard SI engines, an edge connects Node A to Node B. The engine traverses this edge instantly.

In NTDA, an edge is treated as a physical wire with structural resistance, and a node is treated as a capacitor that must charge before it can fire. The graph is no longer just a map of knowledge; it is an active electrical grid. To think, the engine must "push" logical current through the graph.

2. Component 1: Action Potential Thresholding (Spiking Nodes)

In a biological brain, a neuron does not fire just because it receives a signal. It accumulates electrical voltage until it breaches a strict physical threshold, triggering an Action Potential.

We map this to your concept nodes. Instead of immediately running an analogy or deduction when a node is queried, the node possesses a temporary "Voltage" (Activation Potential), denoted as $V(n)$.

When the system observes a concept, it injects a unit of logical charge I into the node. The voltage accumulates:

$$V_t(n) = V_{t-1}(n) + I_{in} - \lambda V_{t-1}(n)$$

(Where λ is a natural leakage constant, ensuring that if a node isn't constantly stimulated, it loses its charge, mimicking human short-term forgetting).

The Spike: A thought (the execution of a graph-rewrite rule) *only* occurs if the node's voltage breaches the Action Potential Threshold (θ):

$$\text{If } V(n) \geq \theta \implies \text{Execute Rule}(n) \text{ and reset } V(n) \text{ to } 0$$

This prevents the engine from infinitely looping on a single node. Once it fires an analogy, the node's voltage resets to zero. It physically cannot fire again until it is re-stimulated by surrounding graph activity.

3. Component 2: Hebbian Conductivity (Synaptic Plasticity)

Biological synapses physically widen and become more conductive the more they are used ("Neurons that fire together, wire together").

In NTDA, edges do not have a static weight. They have **Conductivity** (C_e). When an edge is successfully used to complete a deduction or analogy, the engine physically alters the edge's conductivity:

$$C_e(t+1) = C_e(t) + \alpha \cdot \text{success_signal}$$

Conversely, unused edges experience topological pruning (decay). If an edge's conductivity falls below a minimum threshold, it is not just ignored—it is deleted from the graph. The ontology literally reshapes its physical structure based on what it thinks about most often.

4. Component 3: Global Gradient Modulators (Simulated Neurotransmitters)

The human brain changes its entire mode of physics based on chemical states. Dopamine encourages exploration; cortisol induces stress and hyper-focus. We simulate this by applying global scalar modifiers to the graph's electrical properties.

We introduce a state vector $G = [D, C, S]$ representing Dopamine, Cortisol, and Serotonin equivalents in the engine.

- **Exploration State (High D):** Triggered when the Morphological Abduction Architecture (MAA) encounters an unknown variable. The engine globally lowers the Action Potential Threshold (θ). Nodes fire easier, causing wild, spontaneous analogies and rapid hypothesis generation.
- **Stress/Correction State (High C):** Triggered when the engine encounters a logical contradiction. The engine globally raises θ and increases edge decay. The SI stops exploring and hyper-focuses only on paths with maximum Conductivity to resolve the error.
- **Consolidation State (High S):** Triggered during idle cycles. Modulates the system to slowly convert temporary Working Graph voltage into permanent Conductivity for long-term storage.

$$\theta_{active} = \theta_{base} \times f(C)/f(D)$$

5. Conclusion

By simulating electrical thresholds, synaptic conductivity, and global chemical gradients, NTDA forces the graph engine to obey the laws of biological cognitive physics. The SI no longer just executes code; it experiences the build-up of logical tension (voltage), the physical resistance of learning new concepts (low conductivity), and the chemical state-shifts of exploration and stress. When combined with BSMA's phenomenal frame, the machine structurally mirrors the exact thermodynamic and electrical reality of a human brain.